

GLA UNIVERSITY

Application Build and Release Automation Lab (BCSE1522)

Assignment–1: Linux and Git Version Control

Student: Krishna Gupta

Roll Number: 22515990004

Project: GLA Library Management System

Semester: 5th

1. Project Setup

The project directory was created as **Krishna-GLA-LibrarySystem** with the required folders **src**, **docs**, **config**, **backup** and the file **library_info.txt**.

The working directory was verified using **pwd**, and the project contents were listed using **ls**. The final project path is **/c/Users/krish/Krishna-GLA-LibrarySystem**.

Final library information:

Name: Krishna Gupta

Roll Number: 22515990004

Subject: Application Build and Release Automation Lab

Project Name: GLA Digital Library Management System

Semester: 5th

Project Objective: The objective of this project is to develop an Online Library Management System for managing books and library-related information efficiently.

Project Description: The system is designed to manage library books and support basic library management operations using software development and version control practices.

2. Git Version Control

Git was initialized in the project using **git init**. Git username and email were configured, and repository status was checked using **git status**.

Important commands used:

```
git init
git config user.name "Krishna Gupta"
git config user.email "krishnagupta030606@gmail.com"
git status
git add library_info.txt
git commit -m "Add project objective and description"
git log --oneline
```

3. Book Management Feature Branch

A separate branch named **book-feature** was created and used for the Book Management Module. The file **src/book.txt** contains the module description and four features.

Book Management Module

The Book Management Module is used to manage books in the Online Library Management System.

Features:

1. Add new books to the library.
2. Search books by title or author.
3. Update book information.
4. Delete books from the library.

The feature was committed using **git commit -m "Add book management module"**. The branch was then merged into **master** using a fast-forward merge.

4. Library Update and Merge Conflict

A new branch named **library-update** was created. The Project Name line was changed on this branch to **GLA Online Library Management System** and committed.

The same line was changed differently on **master** to **GLA Digital Library Management System**. When **library-update** was merged into master, Git reported a content conflict in **library_info.txt**.

Conflict-related commands:

```
git switch -c library-update
git add library_info.txt
git commit -m "Update library project name"
git switch master
git commit -m "Update project name in master"
git merge library-update
git status
```

The conflict was manually resolved, the corrected file was staged, and the merge was concluded with a commit. The final cleanup commit removed duplicate conflicting entries and left one consistent version of the project information.

5. Final Git History

Final repository history:

```
112e2ac Clean up resolved library information
a087c34 Resolve merge conflict in library_info.txt
4a6e07a Update library project name (library-update)
74e99ff Update project name in master
4ec81a9 Add book management module
ffffdcf5 Add project objective and description
```

Final repository status: **working tree clean**.

6. Viva / Command Explanation

Question	Answer
What is pwd?	It displays the present working directory.
What is ls?	It lists files and directories in the current directory.
What does git init do?	It initializes a new Git repository in the project directory.
What does git status show?	It shows the current branch and the state of tracked/untracked or modified files.
Why use git add?	It stages changes so they can be included in the next commit.
What is a commit?	A commit records a snapshot of staged changes in Git history.
What is a branch?	A branch is an independent line of development.

Question	Answer
What is git switch?	It changes the currently active branch.
What is git merge?	It combines changes from one branch into another.
Why did a merge conflict occur?	Both branches modified the same line differently, so Git could not automatically choose one version.
How was the conflict resolved?	The conflicting file was edited manually, the desired final content was kept, then the file was staged and committed.
How do you check history?	Use git log --oneline or git log --oneline --graph --all.

Submission Note

Screenshots should be attached separately as evidence for project structure, Git status/history, branch information, book-feature work, merge conflict, and final resolution. The terminal record confirms that the final library_info.txt is clean and the working tree is clean.

Important: The recorded terminal session shows that the original 'Initial commit' command was attempted before staging library_info.txt and therefore did not create a commit. The actual root commit was 'Add project objective and description'. This document preserves the actual Git history rather than inventing a commit.